



GE Medical Systems

Technical Publication

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Revision 1

**Service Note T1439 - Daily IQ Update
in H1.3 eFlextrial Patch Software**

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REVISION HISTORY

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Revision	Date	Reason for change
0		Initial Release
1	09/14/01	Updated procedure, added flowchart, update text

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STATEMENT OF PROBLEM

The LightSpeed QX/i H1.3 eFlextrial Software Patch Release implements some changes in the logic for the Daily IQ Check that may increase the frequency of the following error message in the gesyslog:

The following is an example of the Daily IQ check failure message that may be in the gesyslog:

DAILY IQ CHECK THRESHOLD FAILURE - Scan 50006/6/30

*Drift check fail Kv - KV_120, Sfov - SFOV_LARGE, Colimation - SLICE_COLLIMATION_4x125,
Focal spot - FOCAL_SPOT_LARGE, Das Gain - Value:8,*

*Row - 3, Channel - 442, Baseline mean - 0.184531, Current mean - 0.184589, Smoothed
ratio - 0.000508, Lower limit - -0.000500, Upper limit - 0.000500*

996909205 0 1 Sat Aug 4 07:13:25 2001 230023009

THEORY:

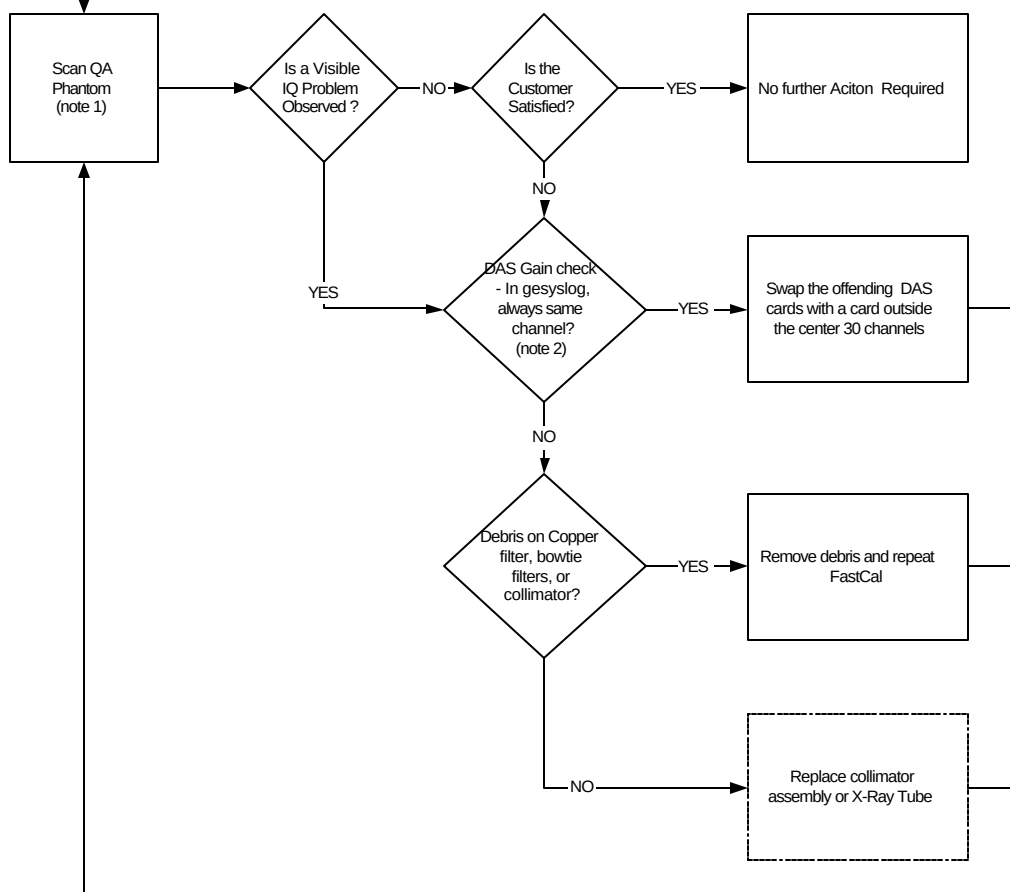
Subtle center spot artifacts (5-20 CT# in depth) are sometimes caused by beam obstructions (grease on tube filter, lead flake, tube port build-up, etc.). When Fastcal is run, the Daily IQ Check test compares the new calibration vectors with the previous day's calibration vector. Only the center 30 channels are analyzed because only these could contribute to a center spot. Errors in the outer channels may result in a band or ring. If the system is operating properly, the new and the old calibration vector should be virtually identical. The Daily IQ Check test looks for changes greater than 0.05% (0.0005).

A failure of the Daily IQ Check test could be due to a lead flake falling on the copper filter, for example, and casting a shadow on some of the center 30 channels. This situation will certainly cause a failure on the first Fastcal after the lead flake is deposited. However, on subsequent Fastcal's the error may or may not occur. On these subsequent Fastcal's the previous day's calibration vector already has the shadow of the lead flake calibrated in. However, the Daily IQ Check test could still fail if the tube temperature during Fastcal is substantially different from the previous day's Fastcal. This would result in a focal spot in a different position and the lead flake's shadow falling on different detector channels.

REMEDIATION:

When this error is displayed in the gesyslog, the following steps should be taken to ensure that image quality is not compromised. For detailed procedures on how to complete these steps, please consult your Service Documentation.

Daily IQ Threshold Failure
Message in gesyslog



Note 1

If the error is observed in the error log, the customer or service representative should run a QA test to verify acceptable image quality for patient scanning. Scan the water section of the QA phantom using the techniques that are failing (shown in the gesyslog). Use the highest mA and slowest scan speed for a quiet image. To most clearly see a possible center artifact due to beam obstruction, wait until the tube is cold. If patients are ever scanned when the tube is cold, perform the QA test with the tube cold. This would result in a worst-case potential IQ scenario. If the results show an image quality issue, it should be addressed.

Note 2

How to Differentiate between DAS drift and beam obstruction - If error consistently occurs on the same channel and the same row, then the error is likely due to DAS gain drift. This scenario may not result in an artifact visible within the customer's QA test.



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